

LECTURE-13

(PENICILLIN: AN ANTIBIOTIC)

➤ ANTIBIOTIC

- literally, ‘antibiotic’ means ‘opposing life’ or ‘against life.’
- a chemical substance produced by a living organism, generally a microorganism, that is detrimental to other microorganisms
- commonly produced by soil microorganisms and that probably represent a means by which organisms in a complex environment, such as soil, control the growth of competing microorganisms
- microorganisms that produce antibiotics useful in preventing or treating disease include the bacteria and the fungi
- in general, *“it may be defined as "antibiotics are compounds produced by bacteria and fungi which are capable of killing or inhibiting, competing microbial species”.*
- **Further,**
 - ✓ an antibiotic is a type of antimicrobial substance active against bacteria
 - ✓ it is the most important type of medicine fighting against bacterial infections, also known as an antibacterial agent
 - ✓ antibiotic medications are widely applied in the treatment and prevention of bacterial infections
 - ✓ antibiotics may either kill or inhibit the growth of bacteria
 - ✓ a limited number of antibiotics have also been found to possess anti-protozoal activity
 - ✓ antibiotics are not effective against the viruses, such as ‘common cold’ or ‘influenza’
 - ✓ drugs which act against viruses are termed as antiviral drugs or antivirals rather than antibiotics
- ✚ before the discovery of antibiotics, many people died from minor bacterial infections, such as 'strep throat', surgery too was risky
- ✚ but after antibiotics became available in the 1940s, life expectancy increased, surgeries got safer, and people could survive what used to be deadly infections
 - ✓ **An ideal antibiotic:** kills or inhibits the growth of pathogenes, causes no allergy to host cells, does not cause damage to host cells, and should be stable when stored in liquid or solid forms
- ✚ **Top 10 classes of antibiotics** are given below. Antibiotics may be available in the form of tablets, capsules, oral suspensions and powder for injection.

The Figure-3 shows antibiotics in various colours, shapes (capsules, tablets) and sizes. The top 10 classes of antibiotics are:

- (1) Penicillins (2) Tetracyclines (3) Cephalosporins (4) Quinolones
(5) Lincomycins
(6) Macrolides (7) Sulfonamides (8) Glycopeptides (9) Aminoglycosides (10) Carbapenems

Classification of Antibiotics

There are various schemes, such as (1) chemical structure, (2) mechanisms of action, (3) spectrum of activity, (4) mode of action, (5) absorbability. Some are discussed below:

✓ **According to Spectrum of Activity**

Antibiotics are classified into (1) narrow or short spectrum, and (2) the wide spectrum

The narrow or the short spectrum antibiotics are those antibiotics which target only specific bacteria and act against narrow range of disease. The examples are: penicillin G, vancomycin, etc.

The broad spectrum antibiotics are those antibiotics which work on many different kinds of bacteria and act against wide range of disease. The examples are: tetracyclin, chloramphenicol, etc.

✓ **According to Mode of Action**

The antibiotics are again two types: (1) the bacteriostatic, and (2) the bacteriocidal.

Bacteriostatic inhibit the growth of bacteria but do not killing them. Example: tetracyclin. Bacteriocidal kills the bacteria. Examples: penicillin, cephalosporin.

✓ **According to Mechanism of Action**

There are many types of antibiotics, like (1) antibiotic inhibiting the synthesis of cell wall, (2) antibiotic interfering with cell membrane integrity, (3) inhibiting nucleic acid synthesis, (4) inhibiting protein synthesis, (5) inhibiting certain metabolic pathway

 antibiotics inhibiting the synthesis of cell wall, example: penicillin

✓ **According to Chemical Structure**

Such as (1) beta lactum, (2) quinolones, sulphonamides, etc.

 Beta lactums are those having beta lactum ring structure in common. This group has the four sub-groups: penicillins, cephalosporins, carbapenems, monobactams

- **The Beta Lactams**

The image of the anti biotic,cephalopirin containing the beta-lactam ring has been shown in the Figure-1, below where the R_1 and R_2 groups are very significant. By changing them various types of cephalosporins may be obtained.

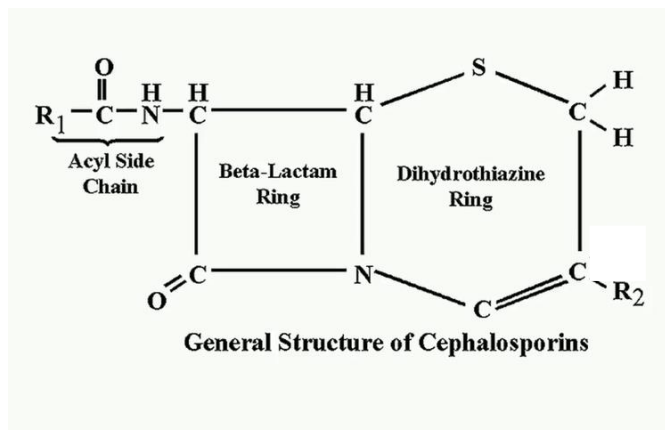


Figure-1: General structure of cephalosporin.

The basic structure of penicillin with the N-acyl group and 6-aminopenicillanic acid (6-apa) has been demonstrated by the Figure-2, below:

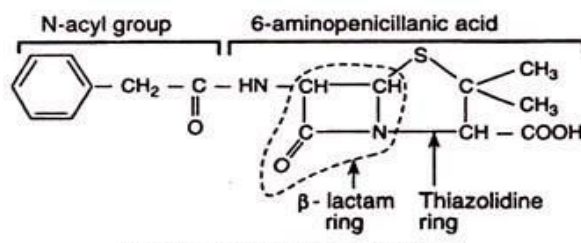


Figure-2: Basic structure of penicillins.



Figure-3: Antibiotics available in various colours, shapes and sizes.

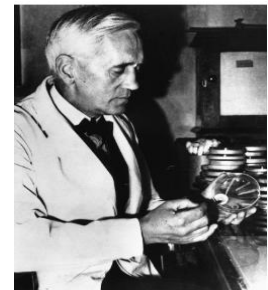
➤ **PENICILLIN**

• **What are penicillins?**

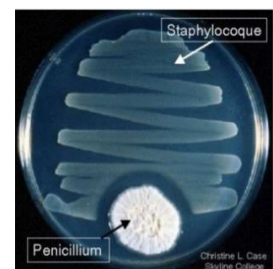
- ✓ penicillins are a group of antibiotics
- ✓ derived originally from common moulds known as *Penicillium* moulds
- ✓ they are antibacterial drugs work against the infections of wide range of bacteria
- ✓ they are bacteriocidal, i.e. they kill the bacteria which cause infections
- ✓ they have beta lactam ring in common, and hence come under the beta lactam group of antibiotic based on chemical structure
- ✓ penicillin works by interfering with bacteria cell walls
- ✓ penicillins revolutionized the history of medical sciences
- ✓ they were the first antibiotic that doctors used and saved millions of life
- ✓ there are several antibiotics in the penicillin class, such as penicillin G, penicillin V, procaine penicillin, and benzathine penicillin

• **What is the history behind the discovery and production of penicillin?**

- ✓ penicillin was the first antibiotic discovered in 1928 by Alexander Fleming, Professor of Bacteriology at St. Mary's Hospital in London
- ✓ the discovery was accidental, and the tale is very interesting
- ✓ returning from holiday on September 3, 1928, he started sorting out the petri dishes containing the *Staphylococcus* culture (which causes boils, soar throats and abscesses)
- ✓ mistakenly, one petri dish was left open, was contaminated
- ✓ he noticed something unusual on the petri dish
- ✓ it was dotted with colonies, save for one area where a blob of mold was growing (evident from the figure)
- ✓ after isolating the sample and testing it, he found that it belonged to the penicillium family, *Penicillium notatum*
- ✓ it was clear, as if the mold had secreted something that inhibited the bacterial growth
- ✓ Fleming found that his 'mold juice' was capable of killing a wide range of harmful bacteria, such as *streptococcus*, *meningococcus* and the *diphtheria bacillus*
- ✓ in the early stage, it was difficult to convince people regarding its uses



Alexander Fleming, holding a petri dish.



Howard Walter Florey
(1898 – 1968)

Ernst Boris Chain
(1896 – 1979)

- ✓ but in 1939, using Fleming's work, two researchers, Howard Flory and Ernst Chain managed to purify penicillin in a powdered form
- ✓ in 1941 they got success in treating a patient
- ✓ in 1943 they produced penicillin on large scale
- ✓ this helped immensely to casualties who had wounds in WWII
- ✓ thus penicillin was hailed a miracle drug and saved millions of life
- ✓ penicillin's colossal effects led to the awarding of the **Nobel Prize** in Medicine and Physiology in 1945 to Fleming, Chain, and Florey
- **What is the general structure of Penicillin, explain?**
 - ✓ penicillins are a group of antibiotics that contain 6-aminopenicillanic acid with a side chain of acyl group attached to it (Figure-1, Figure-2)
 - ✓ the penicillin nucleus, i.e. the beta-lactam ring is the chief structural requirement for biological activity
 - ✓ the side-chain structures (R_1, R_2) determine many of the antibacterial and pharmacological characteristics
 - ✓ penicillins belong to the beta-lactam group of antibiotics
 - ✓ they are the combination of acyl side chain, beta-lactam ring in the middle and thiazolidine ring as evident from the Figure-4, below:

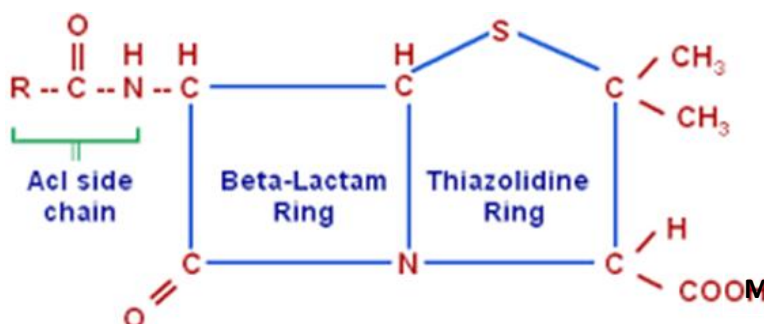


Figure-4: General structure of penicillins.

- ✓ with the variation of R and M in the structure, the type and nature of penicillins vary
- ✓ the significance of R and M may be understood from the following

Table-1:

Derivatives (R in the side chain)	Designation		R in side chain	Units/mg of Na salt
	UK	USA		
2-Pentenyl-	I	F	$CH_3CH_2CH=CHCH_2-$	1600
<i>n</i> -Amyl	Dihydro	Dihydro	$n-C_5H_{11}-$	1500
Benzyl-	II	G	$C_6H_5CH_2-$	1667
<i>p</i> -Hydroxy benzyl	III	X	$HOC_6H_5CH_2-$	900
<i>n</i> -heptyl	IV	K	$n-C_7H_{15}-$	2300
Phenoxy methyl	V	Vee	$C_6H_5-OCH_2-$	1550

- ✓ **Note:** Penicillin is one of the few antibiotics that is still measured in terms of units rather than weight in milligrams or micrograms. One unit of penicillin G represents the specific activity in 0.6 μg of sodium penicillin. Thus 1 mg of penicillin sodium represents 1667 units of penicillin. Similarly other calculations in Table-1.
- ✓ Changes of M in the main ring imparts solubility and ingestion rate control
 - ✚ Direct rapid action injection: M = Na or K
 - ✚ Oral tablets: M = K, Ca, Al
 - ✚ Delayed action oil-based injection: M = procaine or other derivative imparts limited water solubility
- ✓ the basic structure of all penicillins, natural and semisynthetic, is 6-amino penicillanic acid (6-APA) combined with N-acyl group which is variable and shows structural differences in different type of penicillins
- ✓ the N-acyl group is the side chain attached to the amino group of 6-amino penicillanic acid
- ✓ however, there are three natural penicillins that are produced directly and can be obtained from the fermentation liquours of *Pencillium*
- ✓ these are penicillin G, penicillin V and penicillin F
- ✓ natural penicillins are obtained as salts of sodium (Na) or potassium (K) or procaine.
- ✓ penicillins are generally available in the form of sodium (Na) and potassium (K) salts
- ✓ the sodium salts of different penicillins have been shown in the Figure-5.

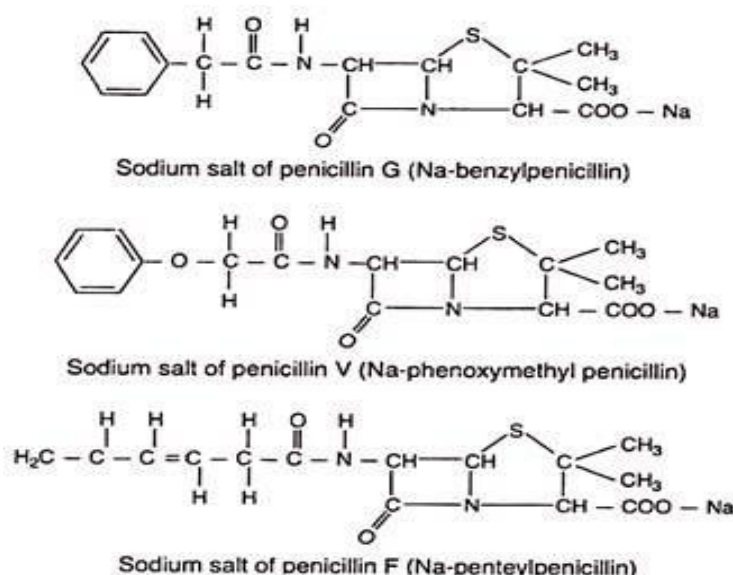


Figure-5: Sodium salts of different penicillins.

- **What is an ideal penicillin?**

An ideal penicillin contains the following properties:

- ✓ kills pathogenic bacteria causing infections
- ✓ causes no allergy to the host cells or the patients on whom administered
- ✓ does not cause damage /harm to the host cells or the patients on whom used
- ✓ should be stable when stored in liquid or solid forms

- **Classification of penicillins based on the mode of production**

There are two broad classification based on the way penicillins are synthesized. They are natural, biosynthetic and semisynthetic.

- ✓ **Natural Penicillins**

- are those produced or harvested naturally by the mold through fermentation
- they were the first antibiotics used in clinical practice
- they inhibit bacterial cell wall synthesis and thus kill or inhibit the propagation of infecting bacteria
- Example: Penicillin G (benzyl penicillin), the addition of precursor, phenoxy acetic acid only stimulates the synthesis of Penicillin G, i.e. enhances the yield

- ✓ **Biosynthetic Penicillins**

- are those synthesized by the mold on addition of the precursor
- Example: Penicillin V (phenoxymethyl penicillin), is produced by the mold only when phenoxy acetic acid is added as a precursor during fermentation

- ✓ **Semisynthetic Penicillins**

- many penicillins cannot be made at all by fermentation, presumably because their side chain acid cannot be taken up by mycelia or cannot be activated once inside mycelia
- such penicillins are manufactured by chemical reactions which attach the side chain to the 6-aminopenicillanic acid (6-APA)
- the process is known as semi-synthesis, whereas the substance is semi-synthetic penicillin
- some of these penicillins can also be made by enzymatic synthesis
- the discovery of penicillin early in this century revolutionized medical care
- although penicillin enjoyed remarkable success against various bacterial agents
- but the emerging bacterial resistance has limited its effectiveness
- the development of the semisynthetic penicillins has prevented the penicillins from becoming obsolete
- the semisynthetic penicillins have improved coverage and effectiveness against a wide range of organisms, including most

streptococcal and staphylococcal species, aerobic gram-negative organisms, and many anaerobic organisms

- the semisynthetic penicillins can be administered either orally or parenterally and in many cases are less costly to administer than other antibiotics
- the semisynthetic penicillins can be used singly in most situations, but when treating highly resistant organisms, such as pseudomonas, combined therapy with an aminoglycoside may be required. In obstetrics and gynecology, the semisynthetic penicillins are used in surgical prophylaxis, intra-amniotic infection, mastitis, endometritis, and other pelvic infections
- Examples are amoxicillin, ampicillin, azidocillin, azolocillin

- **What are the various formulations of penicillins available?**

Penicillins are available in the following formulations:

- ✓ tablets, capsules, oral suspensions, powder for oral suspension, powder for injection

- **What are the examples of penicillin antibiotics available?**

Following are some examples:

- ✓ penicillin V
- ✓ penicillin G (Pfizerpen, Permapen)
- ✓ amoxicillin (Amoxil)
- ✓ amoxicillin/clavulonate (Augmentin)
- ✓ ampicillin (Unasyn)
- ✓ nafcillin (Nallpen)
- ✓ oxacillin (Bactocill)
- ✓ dicloxacillin (Dycill, Dynapen are discontinued brands in the US; generic is available)
- ✓ cloxacillin (discontinued in the US)
- ✓ piperacillin (Pipracil)
- ✓ piperacillin/tazobactam (Zosyn)
- ✓ ticarcillin (Ticar) (Discontinued in the US; ; generic is not available)
- ✓ ticarcillin/clavulonate (Timentin) (Discontinued in the US and a generic is not available.)

- **What are the side effects of penicillin?**

The side effects are:

- ✓ diarrhea that is watery or bloody

- ✓ fever chills, body aches, flu symptoms
 - ✓ easy bruising or bleeding, unusual weakness
 - ✓ urinating less than usual or not at all
 - ✓ severe skin rash, itching, or peeling
 - ✓ agitation, confusion, unusual thoughts or behavior, or
 - ✓ seizure (black-out or convulsions)
- **What is antibiotic resistance?**
 - ✓ antibiotic resistance occurs when bacteria change in some way that reduces or eliminates the effectiveness of drugs, chemicals or other agents designed to cure or prevent infections
 - ✓ the bacteria survive and continue to multiply causing more harm
 - ✓ it is the ability of the microbes to resist the effects of medication that once could successfully treat the patient
- **World Health Organization (WHO) on “antibiotic resistance”**
 - ✓ antibiotic resistance is one of the biggest threats to global health, food security, and development today
 - ✓ antibiotic resistance can effect any, of any age, in any country
 - ✓ antibiotic resistance occurs naturally but misuse of antibiotic in humans and animals is accelerating the process
 - ✓ a growing number of infections – such as pneumonia, tuberculosis, gonorrhea, and salmonellosis – are becoming harder to treat as the antibiotics used to treat them become less effective
 - ✓ antibiotic resistance leads longer hospital stays, higher medical costs and increased mortality

➤ COMMERCIAL PRODUCTION OF PENICILLIN

- Schematic diagram representation of commercial production penicillin

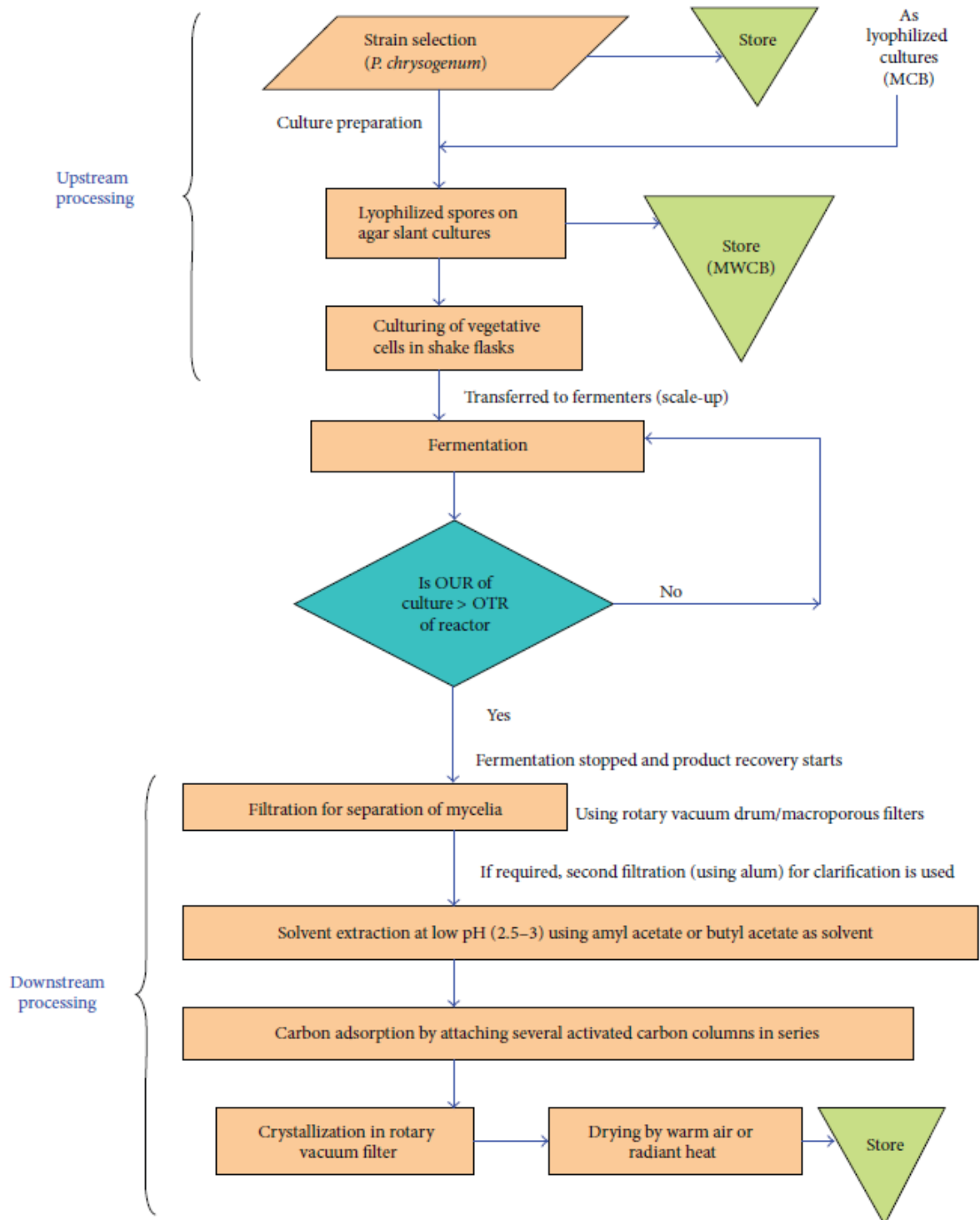


Figure-6: Schematic representation for large-scale production of Penicillin G. Steps are self explanatory "OUR": oxygen uptake rate, "OTR": oxygen transfer rate, "MCB": master cell bank, and "MWCB": manufacturer's working cell bank.

• **Commercial Production: Aerobic Batch Fermentation**

- ✓ Aerobic batch fermentation has been shown in the Figure-7. The culture, *Penicillium crysogennum* developed on the agar slant is aseptically transferred to shake flask for getting suspension of the culture, then from the shake flask aseptically to seed transfer vessel for further growth. Then from there to transfer vessel (seed tank). Desired physicochemical conditions are maintained in these steps. When the seed culture is ready, then it is transferred to the final fermentation vessel as shown the Figure-9.
- ✓ The fermentation medium which contains maize steep liquor or similar base is diluted to a 4 – 5 % solids, fortified with 4 – 5 % lactose solids and nutrients, such as magnesium, zinc and/or sodium sulfate, ammonium acetate, and potassium dihydrogen sulfate in amount of less than 0.2%. Calcium carbonate is added to adjust the pH to 5 – 6. The various types and amount of alkali added depends on the starting liquor.
- ✓ In order to control the yield and type of penicillin desired, organic precursors are also added to the main fermentation unit. For example, type-II penicillin or penicillin G requires phenyl acetic acid precursor.
- ✓ The sterilization of the fermentation medium is highly significant. Moist heat sterilization is carried out at 115 °C for ½ hour. Then the medium is cooled to 20 °C, then inoculated with the specific mold culture, *Penicillium crysogennum* for Type-II penicillin (penicillin G).
- ✓ Sterile air is blown through the fermentor for 4 – 5 days under temperature control of $\pm \frac{1}{2}$ °C. Quality and quantity (yield) of penicillin is maximized by periodic assay of the fermentation broth.
- ✓ Separation of the penicillium mold from rest of the broth is accomplished by vacuum rotary filter. The separated mold is used as a food supplement.
- ✓ Separation of penicillin from the mycelia free broth is achieved by solvent extraction. Suspended solids are removed by continuous filtration at 1 – 2 °C. The clear filtrate or beer is adjusted to pH of 2.5 with dilute phosphoric acid (10% H₃PO₄). Penicillin salt is extracted with amyl acetate.
- ✓ The raffinate or water phase is sent to a solvent recovery still (stripper). The extract is buffered with sodium salt solution to a pH of 7.5 in a continuous mixer and the crude penicillin returns to the aqueous phase. The mixture is centrifuged and the solvent is recirculated.

- ✓ The penicillin is purified by a second acidic extraction, followed by a reversion to a pyrogen-free distilled water solution containing the alkaline salt of the desired element i.e. Na, K, Ca, or Al.
- ✓ The purified aqueous concentrate is separated from solvent in a super centrifuge and then pressurized through a biological filter to remove final traces of bacteria and pyrogens. This solution can be concentrated by freeze drying or vacuum spray drying.
- ✓ Crystalline penicillin salts can be obtained by salting out a saturated solution with a neutral salt containing the cation desired in the final penicillin salt. It can be further purified by recrystallization from organic solvents.
- ✓ The oil-soluble procaine penicillin is made by reacting a penicillin concentrate (20 – 30%) with a 50% aqueous solution of procaine hydrochloride. Procaine penicillin crystallizes from this mixture.

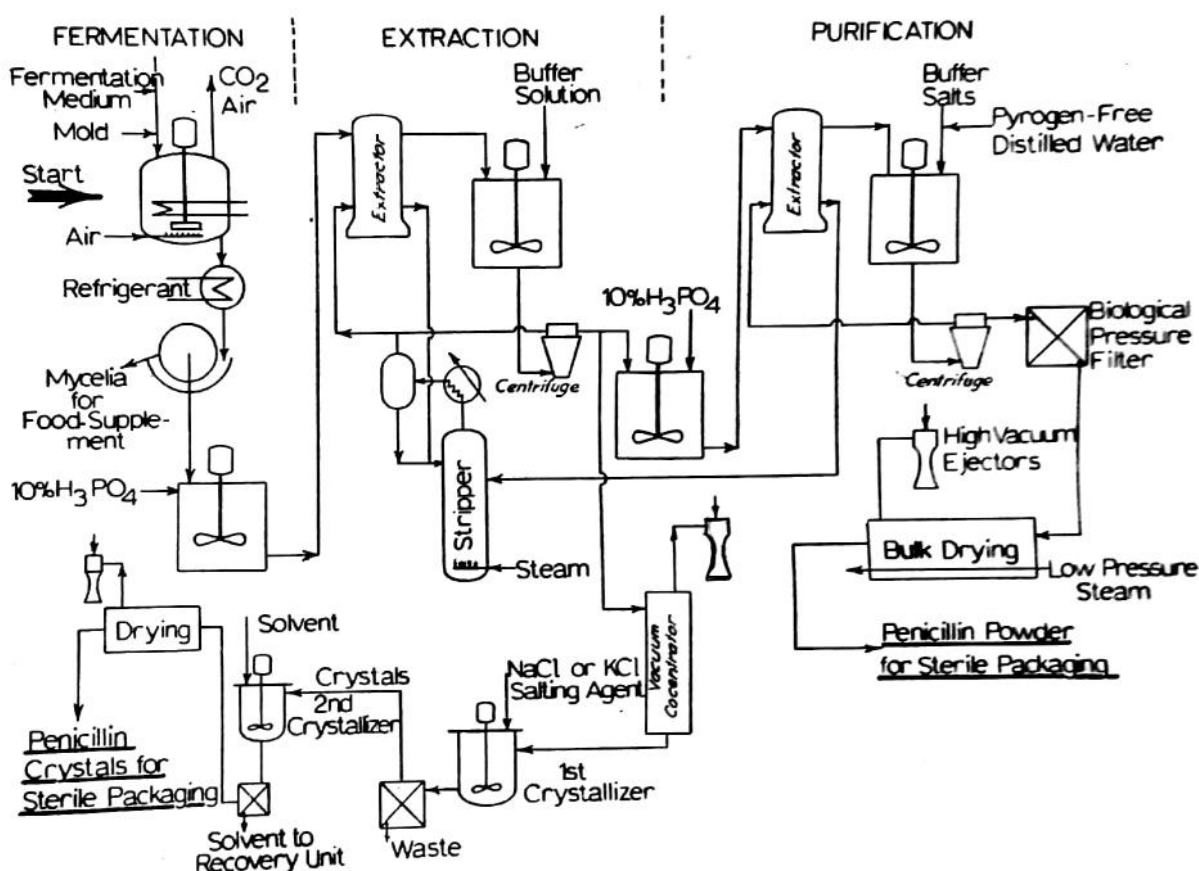


Figure-7: Schematic diagram for batch production of penicillin.

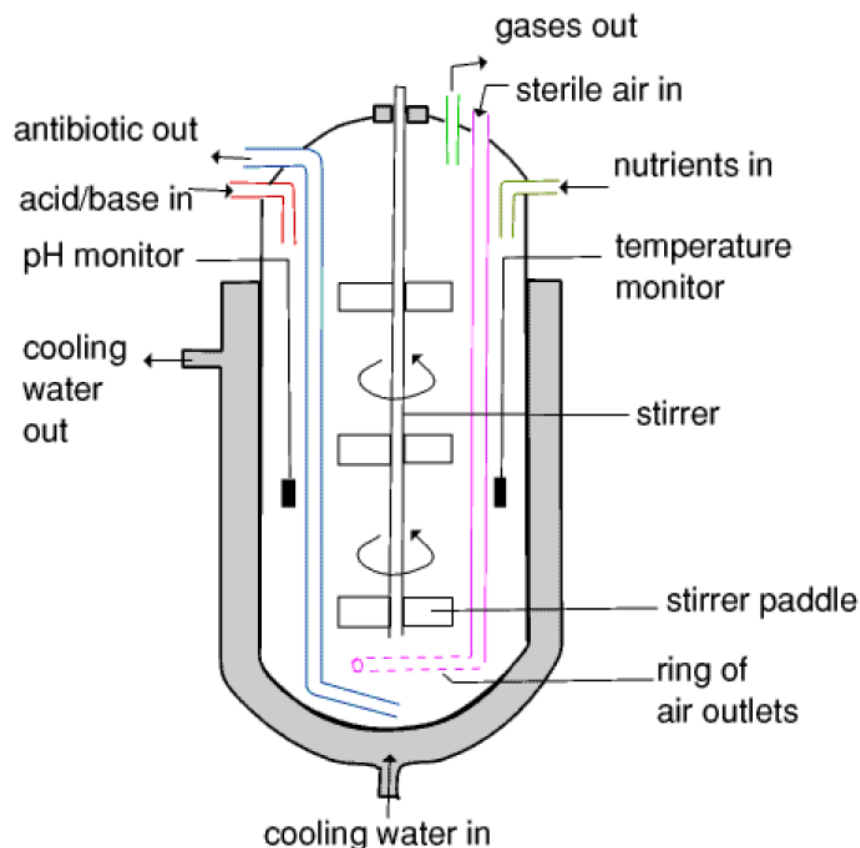


Figure-9: Main fermentation unit (Bioreactor) used for production.

➤ **SIGNIFICANCE OF PRECURSORS IN PENICILLIN BIOSEYNTHESIS**

- ✓ the precursors, ultimately the side chains have great impact on the properties of penicillins
- ✓ if penicillin fermentation is carried out without the addition of side chain precursor, the natural penicillins are formed from which only benzyl penicillin can be isolated
- ✓ however, the desired penicillin can be obtained by adding suitable side chain precursor into the medium
- ✓ such penicillins are called as semi-synthetic penicillins
- ✓ Penicillin-G and Penicillin-V are generally produced commercially
- ✓ when compared to natural penicillins, semisynthetic penicillins have improved characters viz, acid stability, resistance to plasmid or chromosomally coded β -lactamases, expanded antimicrobial effectiveness and are therefore, extensively used in therapy
- ✓ different penicillins with their precursors and R side chains have been shown in **Figure-9**

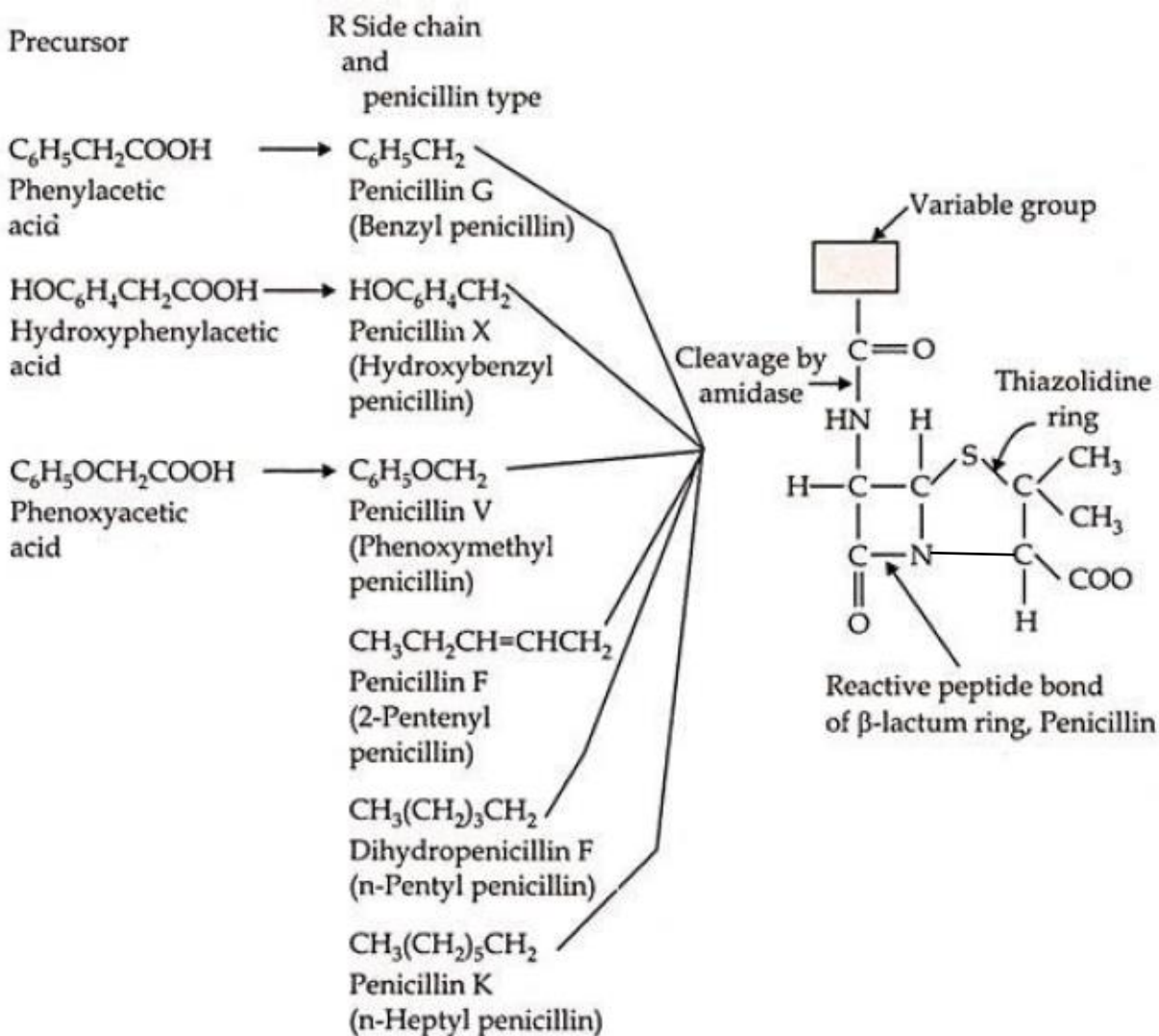


Figure-9: Different penicillins with their precursors and side chain R.